Huzaifah Nadeem

huzaifahnadeem.github.io • h.nadeem@pitt.edu • +1 (412) 785-5086 • Pittsburgh, PA

Education

Ph.D. in Computer Science (Ph.D. Candidate) August 2022 – Present (Expected 2027)
University of Pittsburgh Pittsburgh, PA, USA

- Dissertation (in progress): *Improving Resilience in Industrial Control Systems Using Reconfiguration.*
- Advisor: Dr. Amy Babay.
- Research areas: distributed control systems, formal methods, and quantum networks.

B.S. in Computer Science, Minor in Mathematics 2018 – 2022
Lahore University of Management Sciences (LUMS) Lahore, Pakistan

- Graduated with High Merit; CGPA 3.4/4.0, Major GPA 3.7/4.0; Dean's Honour List.

Research Experience

Graduate Student Researcher September 2026 – December 2026
INfrastructure Sensing for Intelligent Transportation and
Energy Systems (INSITES) Consortium, University of Pittsburgh Pittsburgh, PA

- Project: *Reconfigurable Client Overlay Networks for SCADA Systems.*
- Develops approaches to reconfigure overlay networks for client of a SCADA system.

Graduate Student Researcher September 2024 – August 2026
Cyber Energy Center, University of Pittsburgh Pittsburgh, PA

- Project: *Intrusion Tolerance using Cyber Digital Twins.*
- Investigating how cyber digital twins can strengthen intrusion tolerance in critical-infrastructure control systems, with a particular focus on using them as an upgrade path for legacy systems.

Graduate Student Researcher September 2022 – August 2023
School of Computing and Information, University of Pittsburgh Pittsburgh, PA

- Project: *Severe Impact Resilience: Framework for Adaptive Compound Threats.*
- Studied how to improve control-system resilience against "compound threats," in which cyberattacks are timed to follow a natural hazard (e.g., a hurricane) that has already significantly, though temporarily, degraded the system.

Honors and Awards

Best Paper Award 2024
43rd International Symposium on Reliable Distributed Systems (SRDS 2024)

- For the paper "Tolerating Compound Threats in Critical Infrastructure Control Systems."

SRDS Ph.D. Forum Award 2024
43rd International Symposium on Reliable Distributed Systems (SRDS 2024)

- For the PhD Forum paper on routing protocols for reliable distributed quantum systems.

DSN Student Travel Award 2025
55th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2025)

- Merit-based award supporting participation in the conference.

Full Member 2026

Sigma Xi, The Scientific Research Honor Society

- Nomination by peers to *Full Membership* based on demonstrated noteworthy achievements in peer-reviewed scientific research.

PUBLICATIONS & PRODUCTS

*In the author lists below, my name appears in **bold**; * denotes equal contribution.*

Peer-Reviewed Conference Papers

[C-3] **Availability Under Attack: Quantifying the Impact of Benign Failures on the Resilience of Intrusion-Tolerant Systems**
Aren Alyahya , **Huzaifah Nadeem**, David Tipper, and Amy Babay. **Under Review** at *Network and Distributed System Security (NDSS) Symposium 2027*, Seoul, Republic of Korea

[C-2] **Tolerating Compound Threats in Critical Infrastructure Control Systems**
Sahiti Bommareddy*, Maher Khan*, **Huzaifah Nadeem***, Benjamin Gilby, Imes Chiu, John W. van de Lindt, Omar Nofal, Mathaios Panteli, Linton Wells II, Yair Amir, and Amy Babay. In *Proceedings of the 43rd International Symposium on Reliable Distributed Systems (SRDS)*, Charlotte, North Carolina, September 2024, pp. 66–79. **Best Paper Award.**
DOI: 10.1109/SRDS64841.2024.00017

[C-1] **SoK: A Tale of Reduction, Security, and Correctness – Evaluating Program Debloating Paradigms and Their Compositions**
Muaz Ali, Muhammad Muzammil, Faraz Karim, Ayesha Naeem, Rukhshan Haroon, Muhammad Haris, **Huzaifah Nadeem**, Waseem Sabir, Fahad Shaon, Fareed Zaffar, Vinod Yegneswaran, Ashish Gehani, and Sazzadur Rahaman. In *Proceedings of the 28th European Symposium on Research in Computer Security (ESORICS)*, The Hague, Netherlands, September 2023.
DOI: 10.1007/978-3-031-51482-1_12

Workshop Papers and Other Peer-Reviewed Publications

[W-2] **Using Digital Twins as an Upgrade Path for Critical Infrastructure Control Systems**
Huzaifah Nadeem and Amy Babay. In *1st International Workshop on Digital Twins for Dependability, Resilience and Security (DT4DRS), at the IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, Naples, Italy, June 2025, pp. 168–175.
DOI: 10.1109/DSN-W65791.2025.00057

[W-1] **PhD Forum: Evaluating and Designing Routing Protocols for Reliable Distributed Quantum Systems**
Huzaifah Nadeem. In *Proceedings of the 43rd International Symposium on Reliable Distributed Systems (SRDS)*, Charlotte, North Carolina, September 2024, pp. 330–333. **SRDS Ph.D. Forum Award.**
DOI: 10.1109/SRDS64841.2024.00041

RESEARCH SUPPORT

Non-Monetary Support/Awards

PI, with Co-PI Amy Babay, on National Science Foundation (NSF) ACCESS Discover Allocation **"[CIS262271] Model Checking Reconfiguration Mechanisms for Intrusion-Tolerant SCADA Systems**", 1.5M ACCESS Credits , Oct. 2026 – Apr. 2027.

TEACHING

Instructional Experience

Co-Instructor (with Dr. Amy Babay) Fall 2025
School of Computing and Information, University of Pittsburgh Pittsburgh, PA

- CS 2520 / TELCOM 2321: Wide Area Networks (graduate), covering fundamentals and recent research results.

Guest Lecturer (for Dr. Amy Babay) Spring 2025, 2026
School of Computing and Information, University of Pittsburgh Pittsburgh, PA

- Multiple guest lectures and lab sessions for CS 1652: Data Communication and Computer Networks (undergraduate) and TELCOM 2310: Applications of Networks (graduate), including a lab on socket programming.
- For CS 1652: Data Communication and Computer Networks (undergraduate) and TELCOM 2310: Applications of Networks (graduate). Topics: "Attack-Resilient Application Architectures" and "Advanced Routing".

Teaching Assistant (and *Recitation Instructor) Fall 2023 – Summer 2024
Department of Computer Science, University of Pittsburgh Pittsburgh, PA

- *Fall 2023:* CS 2510 Computer Operating Systems; CS 2520 Wide Area Networks (created two month-long course projects, described below).
- *Spring 2024:* *CS 1613 Quantum Computation, also serving as Recitation Instructor (OMET teaching score 4.50/5.00). The course was being offered for the first time.
- *Summer 2024:* CS 1657 Privacy in the Electronic Society; CS 0007 Introduction to Computer Programming; CMPINF 401 Intermediate Programming.

Undergraduate Teaching Assistant Fall 2020 – Spring 2022
Department of Computer Science, LUMS Lahore, Pakistan

- Teaching Assistant for CS 100 Computational Problem Solving, CS 200 Introduction to Programming, CS 340 Databases, and CS 300 Advanced Programming.
- Prepared and graded labs, assignments, projects, and quizzes, and held office hours and tutorial sessions.

Teaching Credentials

Associate Level Certification in STEM Teaching (ongoing) Expected January 2027
Center for the Integration of Research, Teaching, and Learning (CIRTL), University of Pittsburgh

- Training in research-based teaching practices for STEM courses across different learning environments.

TEACH@Pitt (ongoing) Expected October 2026
University Center for Teaching and Learning, University of Pittsburgh

- Teaching Excellence and Advancement through Certification in Higher Education: training in evidence-based, learner-centered teaching practices.

Achievement in Pedagogy (AiP) Badge 2025
University Center for Teaching and Learning, University of Pittsburgh

- Graduate pedagogy credential, based on structured training in college teaching, earned across three concentrations: Pedagogy, Educational Technology, and Online Teaching.

SERVICE

Program Committees and Reviewing

Program Committee Member, Demo and Poster Tracks
ACM Conference on Computer and Communications Security (CCS) 2026

Shadow Technical Program Committee Member
ACM Internet Measurement Conference (IMC) 2026

Reviewer
37th IEEE International Symposium on Software Reliability Engineering (ISSRE 2026)

Artifact Evaluation Committee Member
IEEE/IFIP DSN 2026, ACM CCS 2026, and NDSS 2027

Conference Support

Student Volunteer 2025
55th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2025)

- Assisted with registration and helped coordinate sessions and social events.

University Service

Mentor Academic Years 2024–25, 2026–27
Graduate Global Ties, University of Pittsburgh Pittsburgh, PA

- Peer mentoring program for incoming international graduate and professional students.

Graduate Student Representative, Academic Council Academic Year 2023–24
School of Computing and Information, University of Pittsburgh Pittsburgh, PA

- Served a one-year term providing graduate-student input on school matters, including the approval of or changes to curricula, academic programs, degree requirements, and admissions policies.